



## Summary

From the literature, we learned that without water-saving basins, the canal chambers would continuously transfer water from upstream to downstream, leading to upstream water scarcity, droughts, and increased risks for navigation. The Panama Canal is a vital trade route, with numerous cargo ships passing through daily to transport goods worldwide. Therefore, our team chose the Panama Canal as the scenario for this study. To prevent droughts, many canals have constructed water-saving basins to reduce the water lost upstream. In this study, we aim to identify strategies for constructing water-saving basins under different scenarios that achieve the highest water-saving efficiency and the lowest construction cost, and we provide recommendations. This offers a mathematical perspective to improve water-saving efforts.

To address the problem of constructing water-saving basins with the highest water-saving rate, we started by analyzing the water exchange diagram between the water-saving basins and the chambers, clarifying the relationships between variables. We then derived the functional relationships between the water-saving basins and the variables. Notably, we represented the unknown trend of the water-saving basin's bottom area using the ratio trend between the basin's bottom area and the chamber's bottom area. Based on this, along with the trends of two other variables, we derived the optimal scale of the water-saving basins. This solves the issue of incomplete utilization of the water-saving potential in basins, which can lead to the unnecessary wastage of water that could otherwise have been preserved, helping to slow the depletion of upstream water resources.

To find the economically optimal parameters for water-saving basins with the highest water-saving efficiency, we considered factors such as energy consumption costs, maintenance costs, water costs, labor costs, and material costs. We combined these costs into an overall cost, built a mathematical model, and integrated it with the parameters derived for the highest water-saving efficiency from the previous step. Using local data and literature, we calculated the annual cost under the economically optimal scenario. This result can help Panama reduce costs, construct larger-scale water-saving facilities, and allocate the saved resources to improving upstream ecosystems or inter-basin water transfers, thereby mitigating water scarcity from multiple perspectives and methods.

In the three-level chamber scenario, the factors and variables considered are closer to real-world conditions, making the problem more complex. Based on the actual operation of water flow, we categorized and analyzed two different operational methods: one with hierarchical connectivity and one without. For each method, we developed mathematical models. Finally, we normalized the data and applied simulated annealing and the branch-and-bound method to obtain results. By comparing these two results, we identified the optimal solution and provided recommendations that are closest to real-world scenarios.

Through this study, we successfully analyzed the scenarios using mathematical modeling to provide the Panamanian government with the most reasonable and efficient recommendations. This, in turn, extends the operational lifespan of the Panama Canal and promotes maritime trade.

**Key words: Panama Canal, water saving basin, MINLP, optimization, cost model, simulated annealing algorithm**



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# 1 Introduction

## 1.1 Background

Shipping, as a crucial mode of trade in today's world, plays a significant role in promoting the development and prosperity of global trade. Large chambers enable connectivity between waterways, enhancing cargo transportation efficiency. However, the operation of canals often faces challenges in balancing economic benefits with water-saving efficiency.

Take the Panama Canal as an example. Despite its narrow terrain and abundant rainfall, Panama struggles to store water effectively, resulting in a shortage of water resources. The canal's operation primarily relies on freshwater from Gatun Lake. Due to the height difference between upstream and downstream chambers, the water consumed by chambers from the upstream is irreversible. Each time a ship passes through the chambers, a large amount of freshwater is drawn from Gatun Lake, leading to its rapid depletion and severely impacting the local economy. Many canals facing similar challenges have built water-saving basins to improve water-saving rates, but balancing the efficiency of these basins with their associated costs remains difficult.

By comprehensively considering the costs of chambers and water-saving basins, we aim to construct a functional model to determine the minimum cost required to ensure water-saving

efficiency. This will provide better construction recommendations to enhance the water-saving rate of chambers.

## 1.2 Problem Restatement

- Determine the function of water-saving rate in relation to the number of water-saving basins, the bottom area of the basins, and their height.
- Construct economic cost models related to the water-saving basins and combine them with the model from the first step to create an overall cost model. Solve for the optimal parameters of this model and calculate the total optimal cost of constructing and operating the basins for one year based on these parameters.
- Further refine the function model under the condition that the chamber has three consecutive chambers and the water-saving basins can exchange water with one another. Develop two models with different scopes of consideration and calculate the water-saving rate and costs for each.

## 2 Variable Table

| Variable Symbol | Meaning                                             | Variable Symbol | Meaning                             |
|-----------------|-----------------------------------------------------|-----------------|-------------------------------------|
| $V_B$           | Volume of water in the water-saving basin ( $m^3$ ) | $i$             | Number of water-saving basin levels |

|             |                                                                                 |              |                                                             |
|-------------|---------------------------------------------------------------------------------|--------------|-------------------------------------------------------------|
| $V_c$       | Volume of a single water-saving basin ( $m^3$ )                                 | $Z$          | Elevation difference of the river ( $m$ )                   |
| $S_B$       | Area of the water-saving basin ( $m^2$ )                                        | $n(x)$       | Number of water-saving basins                               |
| $S_c$       | Chamber area ( $m^2$ )                                                          | $r$          | Water-saving rate (%)                                       |
| $k$         | Ratio of water-saving basin area to chamber area                                | $H_{cl}$     | Height of water not entering the water-saving basin( $m$ )  |
| $H_c$       | Change in water level inside the chamber during each water-saving stage ( $m$ ) | $W$          | Average wage (\$)                                           |
| $H_B(y)$    | Height of the water-saving basin ( $m$ )                                        | $t$          | Time taken to construct the water-saving basin (years)      |
| $C_P$       | Labor cost (\$)                                                                 | $P$          | Number of workers                                           |
| $C_W$       | Cost of wasted water (\$/year)                                                  | $P_W$        | Price of domestic water (\$)                                |
| $V$         | Volume of wasted water ( $m^3$ )                                                | $\eta_W$     | Extraction rate (%)                                         |
| $R$         | Actual water utilization rate (%)                                               | $a(z)$       | Side length of the water-saving basin ground ( $m$ )        |
| $C_{mal}$   | Construction material cost (\$)                                                 | $S_{side}$   | Sidewall area of the water-saving basin ( $m^2$ )           |
| $C_{side}$  | Cost per square meter of water-saving basin sidewall (\$)                       | $C_{bottom}$ | Cost per square meter of water-saving basin bottom area(\$) |
| $k_m$       | Maintenance cost per square meter (\$/year)                                     | $S_{total}$  | Total surface area of the water-saving basin ( $m^2$ )      |
| $C_{en}$    | Energy consumption cost (\$/year)                                               | $E$          | Consumed electric energy (KJ)                               |
| $C_{ele}$   | Cost per kilojoule of electric energy (\$/KJ)                                   | $P_B$        | Power of the water-saving basin valve (%)                   |
| $\eta_p$    | Conversion efficiency of the motor (%)                                          | $T$          | Operation time of water-saving basin gates (year)           |
| $C_{total}$ | Total cost of the chamber                                                       | $C_{maint}$  | Annual maintenance cost(\$/year)                            |

|  |           |  |  |
|--|-----------|--|--|
|  | (\$/year) |  |  |
|--|-----------|--|--|

table 1: Variable Table

### 3 Assumption & Justification

#### 3.1 Assumption

- Assumption: The river in our study has the same flow rate, width, and other characteristics as the Panama Canal, and the environment of the study area shares the same rainfall, sunlight duration, and topography as the Panama Canal.
- Assumption: In the study, the bottom area and height of each water-saving basin are equal.
- Assumption: The data we collected pertains to the period around 2023-2024.
- Assumption: Our study ignores the minor water losses that are inevitable during the operation of chamber gates and cargo transportation processes.

#### 3.2 Justification

- Reason: The canal described in the problem, which requires solutions to water-saving issues, likely faces a water scarcity situation similar to that of the Panama Canal. Therefore, we assume that the study location shares the same geographical and human conditions as the typical Panama Canal.
- Reason: In the actual Panama Canal, water-saving basins are typically constructed with consistent area and height. This facilitates construction management, reduces complexity, and minimizes resource wastage.
- Reason: Specific canal data are significantly affected by historical and climate variations, differing greatly across years. For instance, the Panama Canal underwent expansion in 2016 and experienced a drought in 2023, leading to substantial policy changes. To ensure the accuracy and relevance of the study, we assume the research timeframe is around 2023-2024.
- Reason: In real-world engineering, various minor and uncontrollable factors exist, such as water splashing or adhering to chamber walls during gate operations. Since our model is designed as an idealized computational model, we ignore the unavoidable minor water losses during gate operations and cargo transportation processes.

### 4 Water-saving model

## 4.1 Interpretation

To alleviate water scarcity in upstream arid regions, particularly during the dry season when upstream river flow is insufficient, we propose constructing water-saving basins alongside chambers. This method reduces the proportion of water consumed by chamber operations, minimizing water loss from upstream to downstream during each cycle. Once the basins are constructed, the chambers can operate in a water-saving mode without requiring external power sources such as pumps. The operation of the basins relies solely on gravity and valves.

As shown in Figure 1, when a cargo ship moves from upstream to downstream, the chamber must release water to align its water level with that of the downstream river. During this process, the valve between the chamber and the upstream river is closed, while the valves between the chamber and the water-saving basins open sequentially from top to bottom. A portion of the water in the chamber flows into the basins due to gravity and the height difference between the chamber and the basins. Once the water levels in the chamber and the basins are equal, the valve between the chamber and the downstream river is opened, and another portion of water is released into the downstream river until the water levels are aligned.

As shown in Figure 2, when a cargo ship moves from downstream to upstream, the chamber needs to be filled with water to match the upstream river's level. During this process, the valve between the chamber and the downstream river is closed, and the valves between the chamber and the basins open sequentially from bottom to top. The water previously stored in the basins flows into the chamber due to gravity and the height difference, partially filling its volume. Once the water levels in the chamber and the basins are equal, the valve between the chamber and the upstream river is opened, and the remaining volume is filled with water from the upstream river until the levels are aligned.

In this model,  $H$  represents the height of the water-saving basins,  $n$  represents the number of basins, and  $k$  represents the ratio of the bottom area of the basins to the bottom area of the chamber. Since the bottom area of the chamber ( $A_c$ ) is a constant value for the assumed Panama Canal chamber, we can use  $k$  to reflect the trend (i.e., a larger  $k$  corresponds to a larger basin area). For simplicity, the model derivation initially uses  $k$  to represent this trend, and in the final model, it is substituted back for interpretation.

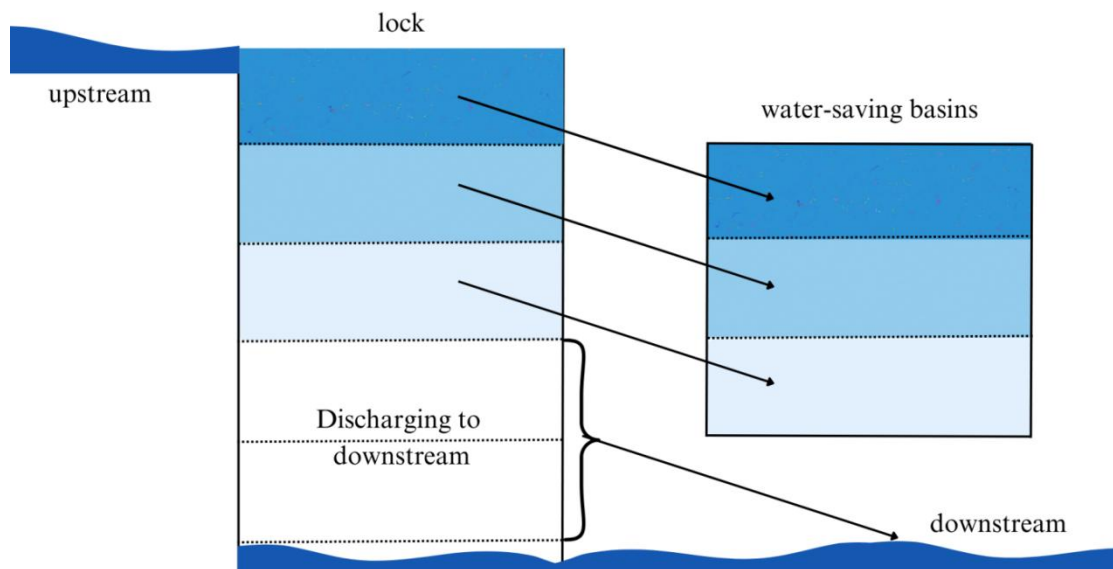


Figure 1

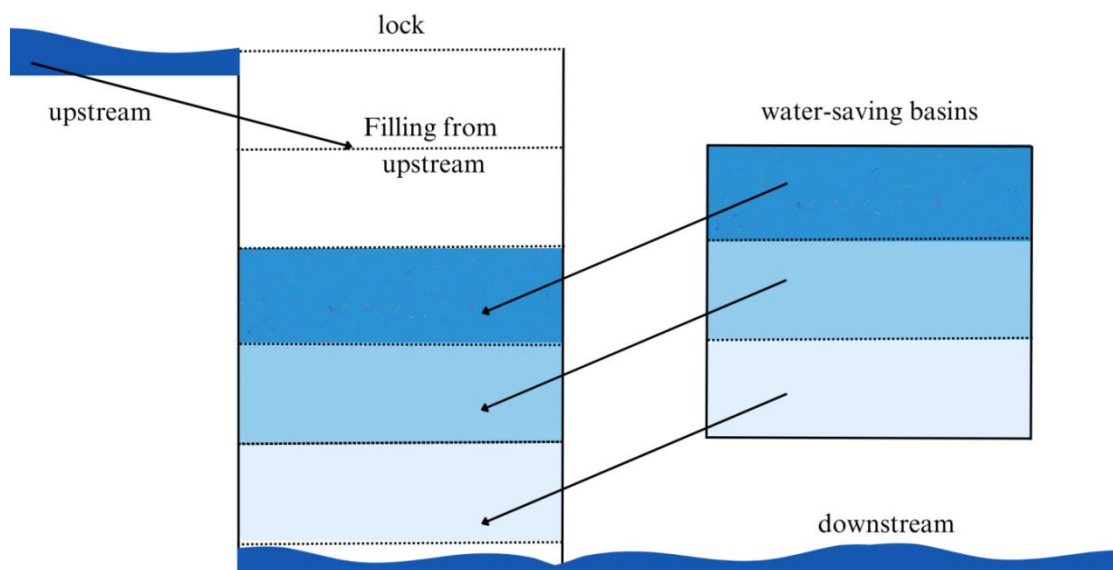


Figure 2

## 4.2 Data collection

We collected data on the newly constructed chambers of the Panama Canal from the official Panama Canal website. As shown in Figure 3, the new chambers of the Panama Canal are 427 meters long, 55 meters wide, and 18.3 meters deep. They can accommodate post-Panamax ships

up to 49 meters wide and 366 meters long, with a carrying capacity of 12,600 twenty-foot equivalent units (TEUs), which is three times the capacity of existing ships.

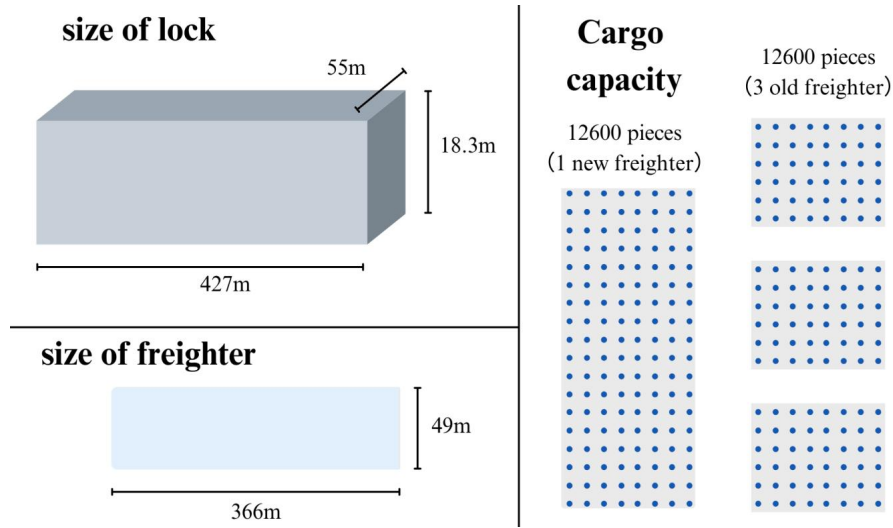


Figure 3

### 4.3 Model

We assume that the bottom area and height of each water-saving basin are equal, and that the volume of water flowing from the chamber into the water-saving basin is equal to the volume of one water-saving basin. Thus, we obtain:

$$V_B = V_C \quad (1)$$

The following formula can be derived from the ratio of the water-saving pool area to the lock chamber area:

$$k = \frac{S_B}{S_C} = \frac{H_C}{H_B} \quad (2)$$

The expression (2) can be rearranged to obtain:

$$H_C = k \cdot H_B \quad (3)$$

Water flows from higher to lower elevations, requiring a height difference, and each cycle requires two instances of this height difference. We assume the most efficient arrangement of the water-saving basins, where the highest point of each basin is not higher than the lowest point of the incoming water, and the lowest point of the basin is not lower than the highest point of the outgoing water, as shown in Figure 4. Additionally, the volume of each portion of water flowing into the water-saving basin is the same, resulting in:

$$H_{cL} = 2H_C = 2 \cdot k \cdot H_B \quad (4)$$

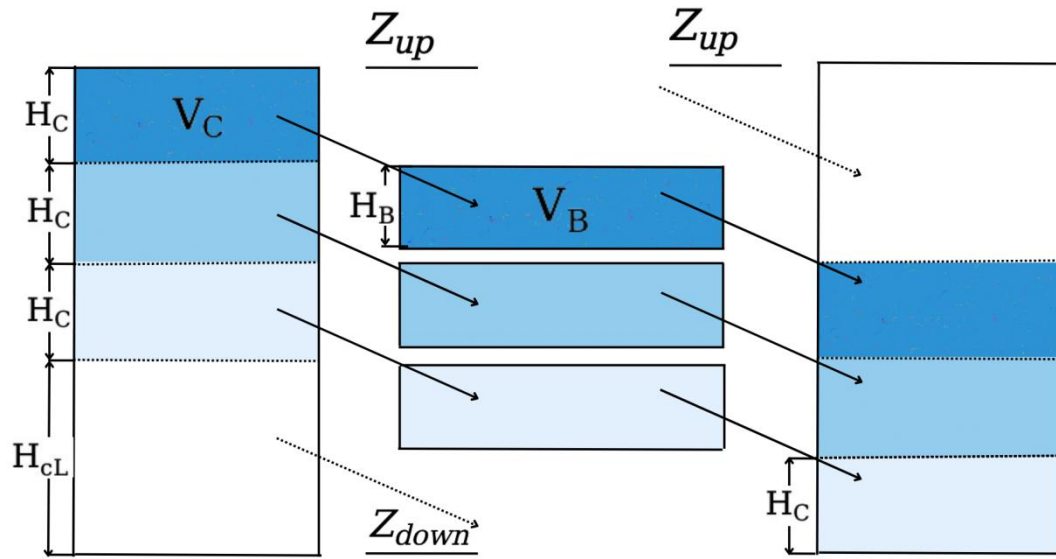


Figure 4

From Figure 4, it can be deduced that the height difference is equal to the elevation difference between the upstream and downstream river, i.e.,  $Z = Z_{up} - Z_{down}$ , and this height difference is also equal to the height of the chamber, resulting in:

$$Z = H_{cL} + \sum_{i=1}^n H_{c,i} \quad (5)$$

Assuming that the height of all the water-saving pools is equal, we get:

$$Z = H_{cL} + n \cdot H_{c,i} = 2 \cdot k \cdot H_B + k \cdot n \cdot H_B = k \cdot (2 + n) \cdot H_B \quad (6)$$

The expression (6) can be rearranged to obtain:

$$H_B = \frac{Z}{k \cdot (2 + n)} \quad (7)$$

The expression (2) can be rearranged to obtain:

$$S_B = k \cdot S_C \quad (8)$$

The water-saving rate formula  $r = \frac{\text{water saved}}{\text{total amount of water}}$  gives:

$$r = \frac{V_B}{V_C} = \frac{H_B \cdot S_B \cdot n}{Z \cdot S_C} \quad (9)$$

Substituting equation (8) into the expression, we obtain the functional relationship between the water-saving rate and the quantity:

$$r = \frac{H_B \cdot k \cdot S_C \cdot n}{Z \cdot S_C} = \frac{H_B \cdot k \cdot n}{k \cdot (2+n) \cdot H_B} = \frac{n}{2+n} \quad (10)$$

Rearranging equation (7), we obtain the relationship between the number of water-saving pools, the height difference, the pool height, and the base area:

$$n = \frac{Z}{H_B \cdot k} - 2 \quad (11)$$

In the formula,  $ZZ$  represents the elevation difference between the upstream and downstream river surfaces, which is a constant. By substituting equation (11) into equation (10), we obtain the functional relationship between the water-saving rate, height, and area:

$$r = \frac{\frac{Z}{H_B \cdot k} - 2}{\frac{Z}{H_B \cdot k}} = 1 - \frac{2 \cdot H_B \cdot k}{Z} \quad (12)$$

Finally, substituting back into the formula, we obtain:

$$r = 1 - \frac{2 \cdot H_B \cdot S_B}{Z \cdot S_C} \quad (13)$$

## 4.4 Result

By establishing an ideal model for the chamber and water-saving basins and making assumptions, we transformed the known data into the number of water-saving basins, area, and height ( $n$ ,  $k$ ,  $H$ ) in various ways, and obtained the functional relationship of the water-saving rate  $r$  with respect to  $n$  and the product.

$$r = \frac{n}{2+n} \quad (14)$$

From the above function of  $r$  with respect to  $n$ , it can be seen that, in the ideal theoretical state, the more water-saving pools there are, the higher the water-saving rate.

$$r = 1 - \frac{2 \cdot H_B \cdot S_B}{Z \cdot S_C} \quad (15)$$

Finally, substituting  $Z$  and  $S_C$  with data values, we obtain:

$$r = 1 - \frac{2 \cdot H_B \cdot S_B}{429775.5} \quad (16)$$

The above equation shows the relationship between  $r$  and the product. In the ideal model, the product and the water-saving rate  $r$  are linearly related. Therefore, as the product (i.e., the volume of the water-saving basins) increases, the water-saving rate  $r$  decreases. Additionally,  $0 \leq \frac{2 \cdot H_B \cdot S_B}{429775.5} \leq 1$ .

## 5 Economic Optimization

### 5.1 Interpretation

In this section, we continue using the Panama Canal as an example. By combining the data we gathered, we have established a total economic cost model for the water-saving basins. The total cost calculated in this model includes both the construction cost and the cost of operating for one year (Figure 5).

During the construction of the water-saving basins, material costs and labor costs are incurred. In the operation phase, energy consumption costs and maintenance costs are generated. Additionally, considering Panama's water scarcity, the model also takes into account the cost of the water saved by the basins (Figure 6).

We have calculated the models for each type of economic cost, and in the final step, we summed them to form the total economic cost model for the water-saving basins. Finally, we used a mixed-integer nonlinear optimization algorithm to optimize the total function, determining the optimal number, height, and bottom area of the basins that minimize the total economic cost, and calculated the minimum total cost for these parameters.

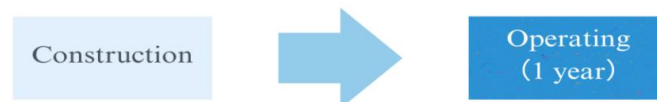


Figure 5

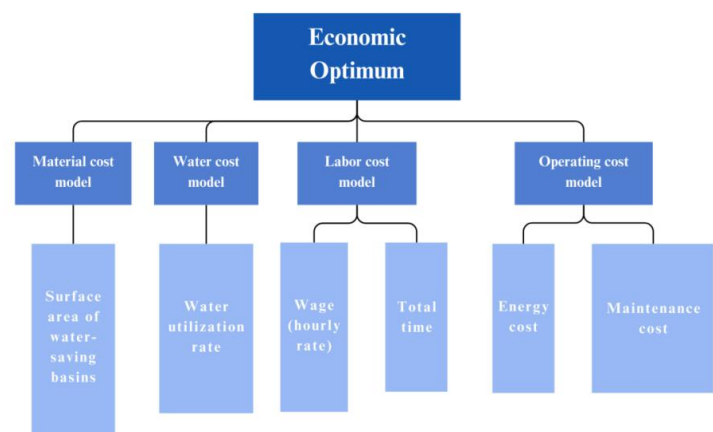


Figure 6

5.2 Data Collection

Regarding labor costs, we found the annual salaries for certain positions of Panama workers on salary websites for 2023, and calculated the hourly wage for each position, as shown in Figure 7.

In a construction team, the proportions of production positions and mechanical supervisors are 45% respectively.

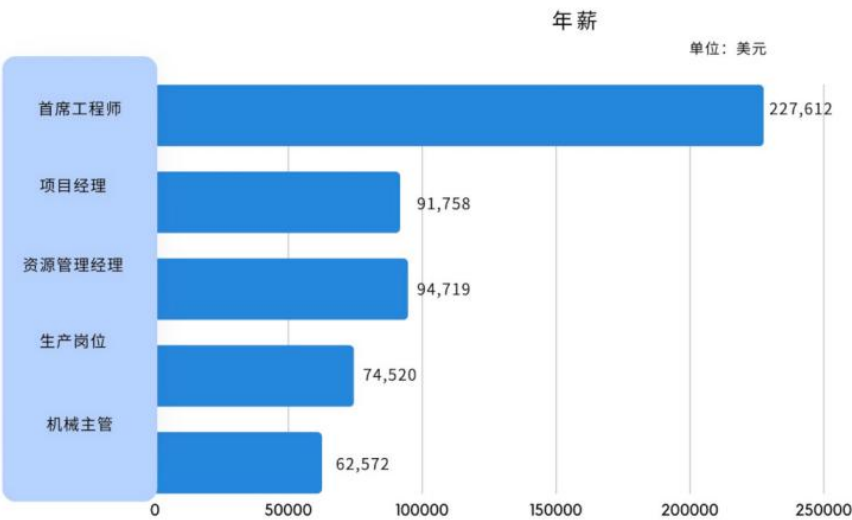


Figure 7

Regarding material costs, we found the prices of some local building materials in Panama on construction material websites, as shown in Table 2:

| Concrete (pool wall)     | Concrete (pool bottom)   | Reinforcement steel | Waterproof materials     | steel         |
|--------------------------|--------------------------|---------------------|--------------------------|---------------|
| 26.50 USD/m <sup>3</sup> | 11.26 USD/m <sup>3</sup> | 792.10 USD/t        | 10.57 USD/m <sup>2</sup> | 2232.30 USD/t |

table 2

And the proportion of different materials used in the construction of water-saving pools according to the official Panama website is shown in Figure 8:

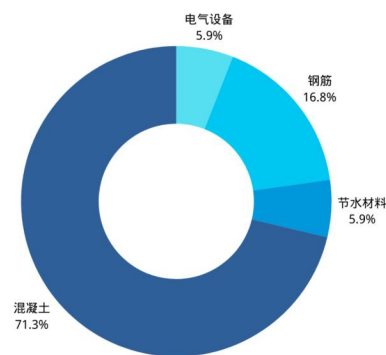


Figure 8. Proportion of water-saving materials

Regarding energy consumption costs, data from the official water energy management website shows that the power consumption is  $P=1\text{kw}$ , with an efficiency of 60%.

Regarding maintenance costs, the World Shipping Trade website provided us with the total maintenance costs for the chambers in 2021. Using this data, we calculated the cost per unit area and multiplied it by the surface area of the water-saving basins to determine the annual maintenance cost for the basins.

Regarding the cost of water consumption, data from the Panama National Website indicates that the water utilization rate is 35%, and the local water rate is \$0.26 per square meter.

## 5.3 Cost Model

### 5.3.1 Labor Cost Model

During the construction of the water-saving basins, labor costs are considered first. Since the bottom area and height of each water-saving basin are equal, meaning the scale of each basin is the same, we assume that all basins are constructed simultaneously in one process. The number of workers required for each basin is the same, and the proportion of different job types is the same. Each worker cannot participate in the construction of two or more basins simultaneously. Using the weighted method, we calculate the weighted average wage ( $\bar{W}$ ), where ( $f$ ) represents the weight, and the formula is as follows:

$$W = \frac{w_1 f_1 + w_2 f_2 + w_3 f_3 + w_4 f_4 + \dots + w_n f_n}{f_1 + f_2 + f_3 + f_4 + \dots + f_n} \quad (17)$$

Since the specific volume of each water-saving pool is not determined, the time is calculated by dividing the water-saving pool volume by the work efficiency, resulting in:

$$t = \frac{V_B}{r} = \frac{H_B \cdot S_B}{r} \quad (18)$$

Based on the previous assumptions and the gathered data, the labor cost formula is derived, where  $P$  represents the number of workers:

$$\begin{aligned} C_P &= n \cdot P \cdot W \cdot t = n \cdot P \cdot W \cdot \frac{H_B \cdot S_B}{r} \\ &= n \cdot 20 \cdot 16 \cdot \frac{H_B \cdot S_B}{16} \\ &= 20 \cdot n \cdot H_B \cdot S_B \end{aligned} \quad (19)$$

### 5.3.2 Materials Cost Model

The construction of water-saving basins consumes a large amount of materials. The materials used and their costs need to be considered during the construction process. In this case, we referred to the official Panama website to obtain the price per square meter of raw materials required for the construction of the bottom and side walls of the water-saving basins.

We assume that only the sidewall area and bottom area of the water-saving basins are considered, ignoring any other materials that might be consumed. We assume the bottom area of the basin remains constant. Let the length of the bottom be (  $a$  ) and the width be (  $b$  ). According to the arithmetic mean-geometric mean inequality, when (  $ab$  ) remains constant, the value of (  $a/b = 1$  ) minimizes (  $a + b$  ). This minimizes the surface area and reduces the material cost.

In this case, (  $a = b$  ), meaning the length and width of the bottom of the water-saving basin are both (  $a$  ), and the bottom area is therefore square. Using the data we obtained, we can derive the material cost formula:

$$\begin{aligned} C_{\text{mal}} &= S_{\text{side}} \cdot C_{\text{side}} + S_B \cdot C_{\text{bottom}} \\ &= n(760 \cdot H_B \cdot a + 80 \cdot a^2) \end{aligned} \quad (20)$$

### 5.3.3 Operating Cost Model

After the construction of the water-saving basins is completed, they begin operation, incurring operational costs. During operation, the basins require regular maintenance, and they also need energy to function. Therefore, we categorize the operational costs into maintenance costs and energy consumption costs.

For maintenance costs, we obtained the price per square meter from the official Panama

website and used this data as a reference. We assume that only the surface of the water-saving basins requires maintenance, and the price of maintenance materials remains constant. Using the data collected, we derived the maintenance cost formula:

$$\begin{aligned} C_{\text{maint}} &= k_m \cdot S_{\text{total}} = k_m \cdot (S_{\text{side}} + S_B) \\ &= 0.37 \cdot n \cdot (4 \cdot H_B \cdot a + a^2) \end{aligned} \quad (21)$$

For energy consumption costs, the opening and closing of the gates are driven by air pumps, which are powered by electricity. We assume that only the electrical energy consumed during the gate operation is considered. For the energy consumption cost, we found the cost per kilojoule of electrical energy  $C_{\text{ele}}$ .  $E$  is the electrical energy consumed,  $n$  is the number of water-saving basins,  $P_B$  is the power of one water-saving basin's valve, and  $\eta_p$  is the conversion efficiency of the motor. Using the data we collected, we derived:

$$\begin{aligned} C_{\text{en}} &= E \cdot C_{\text{ele}} = \frac{n \cdot P_B \cdot T}{\eta_p} \cdot C_{\text{ele}} \\ &= 3333.3n \end{aligned} \quad (22)$$

### 5.3.4 Water Cost Model

The cost of water is defined as the cost of canal water wasted by the chamber. To calculate the cost of the canal water wasted by the chamber, we multiply the volume of wasted canal water that can be utilized by its price. Since the pre-processing water price is unknown, we assume that the local standard water price is used as the price for the actual saved canal water.

Using actual data, we assume that 35% of the water wasted by the chamber is extracted, and 15% of it is actually put to use.

By multiplying the total volume of water wasted each year by  $r$ , we calculate the total volume of canal water that can be utilized from the wasted water by the chamber, and then multiply this volume by the market price of water to obtain:

$$\begin{aligned} C_W &= P_W \cdot V \cdot \eta_W \cdot r \\ &= 0.26 \cdot (Z \cdot S_C - n \cdot H_B \cdot a^2) \cdot 15 \cdot 365 \cdot 35\% \cdot 15\% \\ &= 1.755 \cdot 10^6 - 74.7 \cdot n \cdot H_B \cdot a^2 \end{aligned} \quad (23)$$

## 5.4 Economic Optimal Model

### 5.4.1 Interpretation

By researching data and deriving formulas, we obtained the labor cost model, material cost model, operating cost model, and water cost model.

Since the total economic cost is the sum of the four costs, we can integrate the four cost models to form the total cost model. After obtaining the total cost model, we used a mixed-integer nonlinear optimization algorithm, combined with the derived constraint functions, to calculate the parameter values that minimize the total cost function.

### 5.4.2 Model

The total cost function is equal to the sum of the four cost functions, which gives:

$$C_{\text{total}} = C_{\text{mal}} + C_{\text{maint}} + C_{\text{en}} + C_{\text{W}} \quad (24)$$

After simplifying the above functions, the total cost model is obtained as:

$$f(x) = 3333 \cdot n + 80.37 \cdot n \cdot a^2 - 54.7 \cdot n \cdot H_B \cdot a^2 + 761.48 \cdot n \cdot H_B \cdot a + 1.76 \cdot 10^6 \quad (25)$$

### 5.4.3 Non-Linear Optimization

To facilitate the code execution, in the following text, we will use  $x$  to represent  $n$ ,  $y$  to represent  $k$ , and  $z$  to represent  $a$ .

From equation (25), our objective function is:

$$f(x) = 3333 \cdot x + 80.37 \cdot x \cdot z^2 - 54.7 \cdot z \cdot y \cdot z^2 + 761.48 \cdot x \cdot y \cdot z + 1.76 \cdot 10^6$$

After obtaining the objective function, and combining the previously reviewed data, we get the constraint functions as follows:

$$1 \leq x \leq 10, \text{ and } x \text{ is an integer} \quad (26)$$

$$0 \leq y \leq 7 \quad (27)$$

$$0 \leq z \leq 55 \quad (28)$$

$$0 \leq x \cdot y \cdot z^2 \leq S_c \cdot Z, \text{ which means } 0 \leq x \cdot y \cdot z^2 \leq 429776 \quad (29)$$

$$(x+2)y \leq 18.3 \quad (30)$$

Equations (26) to (28) represent the constraints we have applied to the number, height, and bottom area of the water-saving basins, considering the actual conditions of the canal and the feasibility of construction.

Equation (29) represents the constraint on the volume of a single water-saving basin, meaning the volume of each basin cannot exceed the total volume of the chamber.

Equation (30) is the constraint that the total height of the water-saving basins cannot exceed the elevation difference between the upstream and downstream river surfaces.

We first normalize the three variables. Based on the actual data of the Panama Canal, we determined the ideal point to be  $x = 5$ ,  $y = 3.5$ , and  $z = 27.5$ . At this point, the formula for the total normalization term is:

$$\lambda_1 \cdot (x - x_0)^2 + \lambda_2 \cdot (y - y_0)^2 + \lambda_3 \cdot (z - z_0)^2 \quad (31)$$

In this example, we believe that, in practical situations, the impact of the number of water-saving basins on cost is greater than the impact of their height on cost, and the impact of basin height on cost is greater than the impact of their bottom area on cost. Therefore, we assign weights to  $\lambda$  as follows:

$$\lambda = [\lambda_1, \lambda_2, \lambda_3] = [100, 50, 10] \quad (32)$$

By substituting the ideal point coordinates of the three variables and  $\lambda$  into the equation, we obtain the regularization term as:

$$100 \cdot (x - 5)^2 + 50 \cdot (y - 3.5)^2 + 10 \cdot (z - 27.5)^2 \quad (33)$$

By adding the regularization term to the original objective function, we obtain the final objective function as:

$$3333 \cdot x + 50 \cdot x \cdot z^2 - 30 \cdot x \cdot y \cdot z^2 + 100 \cdot x \cdot y \cdot z + 176 \cdot 10^6 + 100 \cdot (x - 5)^2 + 50 \cdot (y - 3.5)^2 + 10 \cdot (z - 27.5)^2 \quad (34)$$

In the calculation, we iterated through each value of  $x$  and optimized the values of  $y$  and  $z$  using the `fmincon` function, obtaining the global optimal solution.

That is, when  $x = 3$ ,  $y = 3.66$ , and  $z = 55$ , the value of the objective function is 175,535,667.78 USD.

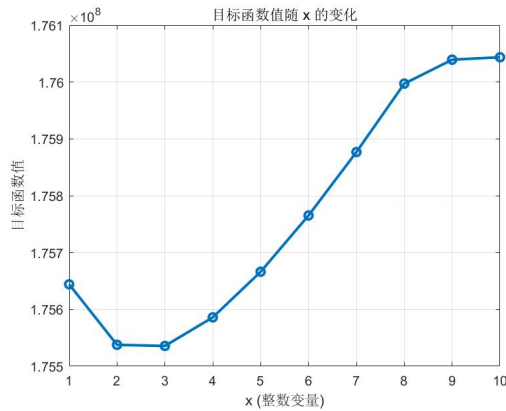


Figure9

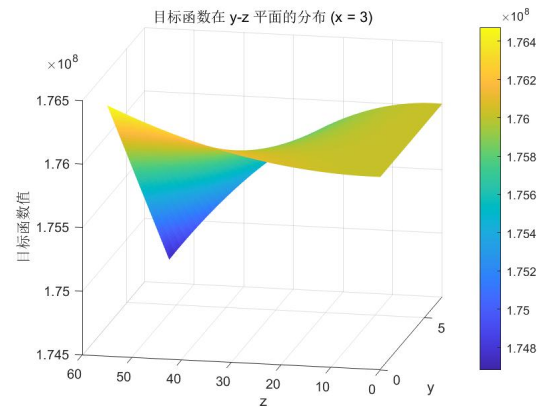


Figure10

## 5.5 Result

After combining the actual data and formula derivations, we obtained the function for each cost. By summing and integrating these cost functions, we derived the total cost function, i.e., the objective function.

Running the objective function, we obtained the parameters that minimize the total economic cost:

$$x=3, y=3.66, z=55$$

That is, the number  $n=3$ , the height  $H_B=3.66$  m, and the bottom area  $S_B=a^2=55^2=3025$  m<sup>2</sup>

When these parameters are input into the objective function, the minimum value of the function is obtained, which corresponds to the minimum total economic cost of 175,535,667.78 USD.

## 6 Three level lock model

### 6.1 Interpretation

Regarding the three-level lock model, we continue to assume the scenario of the Panama Canal, where the bottom area and height of both the water-saving basins and chambers are equal. In addition, considering the influence of water pressure in multi-level chambers, but since the effect of water pressure on the study variables is minimal, we ignore the water pressure and assume that the elevation difference in the three-level chamber is the same to simplify the mathematical model.

Next, we propose three different scenarios for the three-level lock: no water exchange between the basins in the three chambers, water exchange without considering the return of water from the basins to the chambers, and water exchange with consideration of the return of water from the basins to the chambers.

For these three scenarios, we develop mathematical models for each, and apply multivariable optimization and classification methods to calculate their costs and water-saving rates. Finally, by comparing the costs and water-saving rates of the three scenarios, we select the optimal solution and adjust the water-saving strategy.

## 6.2 Basin-Same Level Chamber Model

### 6.2.1 interpretation

In the first model, we assume that the water in the chamber exchanges only with the water-saving basins on the same level. Under this condition, the total amount of water consumed during operation equals the volume of each chamber minus the volume of the water-saving basin multiplied by three times the number of basins.

Based on the data and actual conditions we collected, we imposed constraints on the number, area, and height of the water-saving basins. Using the simulated annealing algorithm, we introduced randomness to escape local optima and gradually reduced this randomness with each iteration, ultimately converging to the global optimum, thus deriving the target model.

### 6.2.2 Model

First, we assume that the volume of water lost by the three locks is equal, then we get:

$$V_1' = V_2' = V_3' \quad (35)$$

Since the amount of water lost after the establishment of the water-saving basins is the total volume of water in the chamber minus the total volume of water in the basins, we can establish the functional relationship with the variables, resulting in the objective function:

$$V_{\text{waste}} = V_1' = V_2' = V_3' = 3 \cdot (Z \cdot S_c - n \cdot S_B \cdot H_B) \quad (36)$$

For the constraint function, we applied the equivalence relation from the second question:

$$H_C = k \cdot H_B = \frac{S_B \cdot H_B}{S_c} \quad (37)$$

Based on the height difference between the water in the water-saving basins and the chamber mentioned in the first question, it can be concluded that the total water level height in the chamber is composed of the heights of two water-saving basins and the actual wall height between the basins. Therefore, we can obtain:

$$n \cdot H_B + 2 \cdot H_C \leq Z \quad (38)$$

Substituting equation (38) into equation (37) gives:

$$n \cdot H_B + \frac{2 \cdot H_B \cdot S_B}{S_C} \leq Z \quad (39)$$

By subtracting the previously estimated water cost model from equation (25), we obtain the individual economic cost model:

$$C_{\text{total}} = 3333 \cdot n + 80.37 \cdot n \cdot S_B + 20 \cdot n \cdot H_B \cdot S_B + 761.48 \cdot H_B \cdot \sqrt{S_B} \quad (40)$$

### 6.2.3 optimization

Objective function of waste water:

$$V_{\text{waste}} = 3(S_C \cdot Z - n \cdot S_B \cdot H_B) \quad (41)$$

Substitute data, we get:

$$V_{\text{waste}} = 1.2893 \cdot 10^6 - 3 \cdot n \cdot S_B \cdot H_B \quad (42)$$

constraint function:

$$(n+2) \cdot H_B \leq 18.3 \quad (43)$$

$$0 \leq n \cdot H_B \leq 18.3 - \frac{S_B \cdot H_B}{23485} \quad (44)$$

We first normalized equation (40) and the objective function:

Based on practical conditions, we defined the range of values for the three variables:

$$\begin{cases} 0 \leq S_B \leq 3025 \\ 0 \leq n \leq 7 \\ 0 \leq H_B \leq 7 \end{cases} \quad (45)$$

According to the normalization formula, we map the function to the interval [0, 1]:

$$C_{\text{total-normalized}}(n, S_B, H_B) = \frac{C(n, S_B, H_B) - C_{\min}}{C_{\max} - C_{\min}} \quad (46)$$

$$V_{\text{waste-normalized}}(n, S_B, H_B) = \frac{V(n, S_B, H_B) - V_{\min}}{V_{\max} - V_{\min}} \quad (47)$$

Then, we assigned weights to them, with the weight ratio of  $w_c:w_v=0.6:0.4$ , based on the current market value judgment of the cost and the water-saving amount.

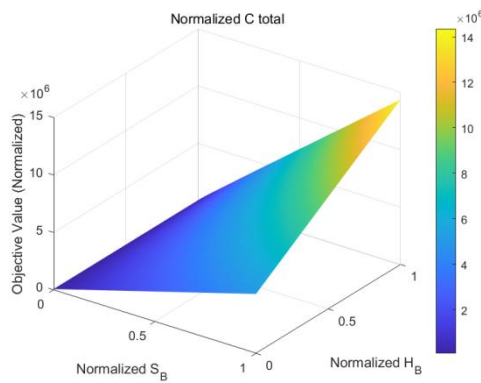


Figure 11

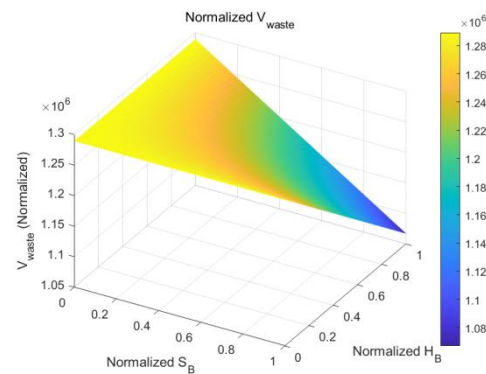


Figure 12

Then, we sum these two functions weighted accordingly to obtain the final overall objective function:

$$f_{\text{total}} = 0.6 * C_{\text{total-normalized}}(n, S_B, H_B) + 0.4 * V_{\text{waste-normalized}}(n, S_B, H_B) \quad (48)$$

In the optimization of the model, we chose the simulated annealing algorithm and performed up to 1000 iterations. Ultimately, we obtained the optimized results. We also adjusted the constraints within a reasonable range and, after conducting the experiment, compared the results to determine the optimized values achievable using this model.

| Parameter      | Best parameter |
|----------------|----------------|
| $n(x)$         | 4              |
| $H_B(y)/m$     | 3.05           |
| $S_B(a^2)/m^2$ | 47.4373        |

Finally, the optimized parameters were substituted into equation (40) to calculate the economic cost of construction and operation for one year:

$$\begin{aligned}
 C_{total} &= 3333 \cdot 4 + 80.37 \cdot 4 \cdot 47.44 + 20 \cdot 4 \cdot 3.05 \cdot 47.44 + 761.48 \cdot 3.05 \cdot \sqrt{47.44} \\
 &= 549712\$ \quad (49)
 \end{aligned}$$

### 6.2.4 result

The optimal parameters calculated by this method and their economic costs are as follows:

| n(x) | HB/m | SB/m <sup>2</sup> | $C_{total}/ \$$ | $V_{saved}/m^3$ |
|------|------|-------------------|-----------------|-----------------|
| 4    | 3.05 | 47.4373           | 549712          |                 |

## 6.3 Basin- Different Level Chamber Model

### 6.3.1 interpretation

First, we assume that the chambers at each level and the water-saving basins, as well as the basins and the chambers, can fully exchange water. By referencing additional data on the principles of three-level chambers, the previously established model is made more complex. After introducing new variables, we use numbers and subscripts to represent and calculate them accordingly.

During the modeling process, the simulated annealing algorithm was applied to optimize multiple new variables. Additionally, the branch-and-bound method was used to perform the primary integer programming.

### 6.3.2 Model

To calculate the volume of water wasted after passing through a lock with water-saving basins, we denote the volume of water wasted per lock level without water-saving basins as  $V_1, V_2, V_3$ . Assuming that the volume of water originally wasted per lock level is equal, i.e.,  $V_1 = V_2 = V_3 = V$ , all subsequent references to the volume of water wasted per lock level without water-saving basins will be represented as  $V$ .

Since there are transitional relationships between the water-saving basins of different lock levels, the number of water-saving basins varies for each level. Let the number of basins for each level be  $n_1, n_2, n_3$ , respectively.

As the total maximum height of the water-saving basins at each level must be less than or equal to the height of the lock at that level, we have the following constraint:

$$0 \leq n_1 \leq \frac{Z-H_{C1}}{H_B} \quad (50)$$

$$0 \leq n_2 \leq \frac{Z-H_{C2}}{H_B} \quad (51)$$

$$0 \leq n_3 \leq \frac{Z-H_{C3}}{H_B} \quad (52)$$

In this case, to ensure that the quantities of each level are not necessarily equal, we assume that the quantities between different levels are unequal,  $H_{C1} \neq H_{C2} \neq H_{C3}$

In this model, we consider both the mutual transfer of water between the water-saving pools and the locks, as well as the mutual transfer between the locks and the water-saving pools. Let the transportation volume between different levels of water-saving pools ( $i_1 \neq i_2$ ), or between locks and water-saving pools at the same level ( $i_1 = i_2$ ), be denoted as  $l_{i_1 i_2}$ , for example, the transportation volume of water from the water-saving pool of the first-level lock to the water-saving pool of the second-level lock is  $l_{12}$ .

Let the transportation volume of water from the water-saving pool at level  $i_1$  to the lock at level  $i_2$  be denoted as  $u_{i_1 i_2}$ , for example, the transportation volume of water from the water-saving pool of the first-level lock to the water-saving pool of the second-level lock is  $u_{12}$ . As shown in Figure 13, by analyzing the situation, we obtain the following results:"

In this model, we consider both the water transfer between the water-saving basins and the chambers, as well as the transfer between the chambers and the water-saving basins. Let the transfer volume between basins of different levels  $T_{ij}$  or between basins and chambers of the same level  $T_{ii}$  be  $T_{ij}$ . For example, the transfer volume from the water-saving basins of the first-level chamber to the basins of the second-level chamber is  $V_{12}$ .

Let the transfer volume between the basins of level  $i$  and the chambers of level  $j$   $T_{ij}$  be  $V_{ij}$ . For example, the transfer volume from the water-saving basins of the first-level chamber to the second-level chamber is  $V_{12}$ . As shown in the Figure, the analysis yields the following results:

$$V_1' = V_1 - l_{11} + u_{11} + u_{21} + u_{31}$$

$$V_2' = V_2 - (l_{22} + l_{21}) + (u_{22} + u_{32})$$

$$V_3' = V_3 - (l_{33} + l_{32} + l_{31}) + u_{33}$$

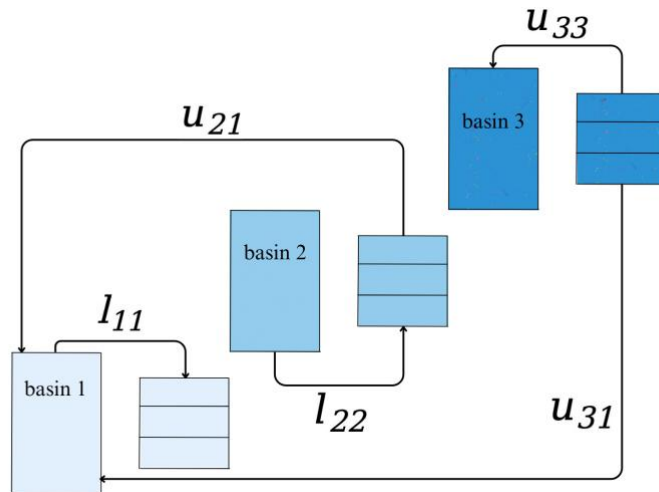


Figure 13

Since the absolute transportation volume of water is smaller than the maximum raw water waste volume at the same level  $V$ , we conclude that:

Since the transferred water volume is absolutely less than the maximum original water waste volume  $V$  at the same level, we obtain:

$$\begin{aligned} l_{11} - (u_{11} + u_{21} + u_{31}) &\leq V_1 \\ l_{21} + l_{22} - (u_{22} + u_{32}) &\leq V_2 \\ l_{31} + l_{32} + l_{33} - u_{33} &\leq V_3 \end{aligned}$$

and

$$\begin{aligned} l_{11} + l_{21} + l_{31} &\leq n_1 \cdot S \cdot h \\ l_{22} + l_{32} &\leq n_2 \cdot S \cdot h \\ l_{33} &\leq n_3 \cdot S \cdot h \end{aligned}$$

Let the volume of water still wasted after the water-saving process in the  $i^{\text{th}}$  level's water-saving basins be  $V_{\text{waste},i}$ , We then obtain:

$$V_{\text{waste}} = V_1' + V_2' + V_3'$$

### 6.3.3 optimization

Wasted water objective function

Water consumption objective function:

$$V_{\text{waste}} = V_1' + V_2' + V_3'$$

Substitute, then we get:

$$\begin{aligned} V_{\text{waste}} &= V_1 - l_{11} + u_{11} + u_{21} + u_{31} + V_2 - (l_{22} + l_{21}) + (u_{22} + u_{32}) + V_3 \\ &\quad - (l_{33} + l_{32} + l_{31}) + u_{33} \end{aligned}$$

Constraint function:

(See the equation above)

We normalized the water consumption objective function:

$$V_{\text{waste-normalized}}(n, S_B, H_B) = \frac{V(n, S_B, H_B) - V_{\min}}{V_{\max} - V_{\min}}$$

The normalized economic cost model here is the same as the previous scheme.

Therefore, the total objective function here is:

$$f_{\text{total}} = 0.6 * C_{\text{total-normalized}}(n, S_B, H_B) + 0.4 * V_{\text{waste-normalized}}(n, S_B, H_B)$$

We applied the simulated annealing algorithm to perform multivariable optimization of this function, while also using the branch-and-bound method. First, we branched the integer variables  $n_1, n_2, n_3, n_4$  by decomposing them into two branches:  $n \leq 4$  and  $n > 4$ , continuously generating new subproblems. Using the simulated annealing algorithm, we estimated the lower bounds for different sub-branches and updated the upper bounds based on the feasible solutions that appeared. We iteratively pruned the remaining solution space, searching for and retaining the optimal solution.

Finally, we solved for the optimal value of the function:

|                  |          |
|------------------|----------|
| $n_1$            | 4.0      |
| $n_2$            | 6.0      |
| $n_3$            | 9.0      |
| $S_B \text{m}^2$ | 17470.14 |
| $H_B/\text{m}$   | 1.5124   |

According to our results, in the three-lock problem, the optimal economic cost and water savings are achieved when 9 water-saving pools are built at the upstream lock, 6 at the middle lock, and 4 at the downstream lock.

From our results, it can be concluded that in the three-chamber lock problem, the optimal economic cost and water-saving rate are achieved when 9 water-saving basins are constructed at the upstream chamber, 6 basins at the middle chamber, and 4 basins at the downstream chamber.

$$V_{\text{saved}} = n \cdot S_B \cdot H_B = 501214/\text{cycle}$$

Because  $V_{\text{saved}}$  of the second arrangement is extremely higher than the first one, so from the aspect of ecology, we prefer the second arrangement which is:

The upstream chamber is equipped with 9 water-saving basins, the middle chamber with 6 basins, and the downstream chamber with 4 basins. Additionally, the bottom area of each basin is  $17,470.14 \text{ m}^2$ , and the height is  $1.512 \text{ m}$

## 7. the strength and weakness of model

In the first question, we derived the relationship between the water-saving rate and the number, height, and area of the water-saving basins during the off-season. The relationship is intuitive from our formula, and since the formula is relatively simple, it is also easy to calculate. From a physical perspective, the formula reflects comprehensive considerations. The units of each variable and constant align with physical significance, and the units of the final result also correspond to the practical meaning of the water-saving rate.

However, the drawback is that our model is overly idealized and cannot be widely applied in practical production scenarios. Forcing its application may lead to errors during construction.

In the second question, we derived the economic optimization model for the water-saving basins in a single chamber lock. Our objective function integrates multiple variables in a nonlinear combination, which effectively captures the trade-offs and relationships within the complex system. The objective function also involves several nonlinear terms and interaction terms, making it suitable for multi-variable optimization problems that require precise modeling, in line with the practical problem we are solving. The constraints we used are clear, with well-defined variable ranges (both lower and upper bounds), effectively avoiding invalid solution regions. The combination of nonlinear inequality constraints keeps the problem within a feasible solution space. We chose an appropriate optimization method, using the Sequential Quadratic Programming (SQP) algorithm, which is well-suited for solving continuous and nonlinear constrained optimization problems.

The drawbacks are that our objective function may be too complex, with higher-order and interaction terms, which could complicate gradient calculations during the optimization process and increase convergence difficulty. The large coefficient differences may cause numerical stability issues, requiring further normalization. Our initial point selection might not be robust enough, as the initial point was arbitrary. If it is far from the global optimal solution, it may lead the algorithm to converge to a local optimum.

In the third question, we derived two models for the water-saving basins in a three-level chamber lock. In the first model, we used the simulated annealing algorithm. Simulated annealing has strong global optimization capabilities, making it suitable for complex nonlinear problems, especially those prone to getting stuck in local optima. The algorithm has global search characteristics and is less dependent on the initial point, which is suitable for optimization problems with a large variable range or complex constraints. Additionally, by introducing a regularization penalty term, we enhanced the model's ability to handle constraints. This penalty term transforms the nonlinear constraint conditions into part of the objective function, making it easier to handle constraint issues in unconstrained optimization algorithms.

However, the drawbacks include insufficient handling of the variable range in the function model. The lower bound  $b$  limits  $x$  (an integer variable), but the ranges for  $y$  and  $z$  were not explicitly defined, and the upper bound  $u_b$  was only set for  $x$ . The lack of a complete range could lead to

ineffective search. We also did not consider integer constraints, as  $x$  needs to be an integer, but simulated annealing by default searches within the continuous domain, which may result in solutions that do not meet the requirements. The model's algorithm efficiency is relatively low, as simulated annealing takes longer to converge compared to gradient-based optimization algorithms, especially when the maximum number of iterations (such as 1000) is set. Parameter tuning is costly, as the initial temperature, cooling rate, and iteration count require repeated testing, increasing debugging time. Additionally, our objective function is relatively simple, containing only one main optimization goal, which is much less complex than the constraints. If the nonlinearity of the objective function is low, it may not fully leverage the global search capabilities of simulated annealing.

## 8.conclusion

To find the optimal water-saving basin size and balance the economic costs to address the upstream drought and water shortage conditions in some canals, we developed four mathematical models that cover physical, historical, economic, and social factors. Ultimately, we derived the functional relationships and optimal recommendations for each question.

By establishing a water-saving rate model, analyzing the transportation process between the chamber and the water-saving basins in a single-chamber lock, and calculating the water-saving rate, we concluded that in an ideal state, the more water-saving basins there are, the higher the water-saving rate, and the larger the volume of the basins  $S$ , the lower the water-saving rate ( $\propto \frac{1}{S}$ ). This laid the foundation for selecting the optimal construction plan for water-saving basins in subsequent research and provided effective recommendations for preventing upstream water scarcity.

After reviewing a large amount of economic and social literature, we gained insights into the costs related to labor, materials, operation, and water consumption for the Panama Canal. Based on this data, we formed cost models for each aspect and performed calculations. After summing these models, we derived the final cost model for the Panama Canal. By integrating this model with the water-saving rate model established earlier, we found that for a single-chamber lock with three water-saving basins, a height of 7.66 meters, and an area of 55, the objective function value is 175,535,667.78 USD. Under these parameters, the lowest cost and highest water-saving rate can be achieved.

Building on the single-chamber lock, we increased the complexity of the variables and applied them to a three-chamber lock. Based on previously collected data and materials, we classified and discussed two scenarios, calculating the optimal solutions for each. The first method assumes no water exchange between the water-saving basins at different levels, and we obtained a result. The second method assumes water exchange between the water-saving basins at different levels, and we obtained another result. Finally, by comparing the two, the best recommendation is the upstream chamber is equipped with 9 water-saving basins, the middle chamber with 6 basins, and the downstream chamber with 4 basins. Additionally, the bottom area of each basin is 17,470.14 m<sup>2</sup>, and the height is 1.512 m.

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